

BTP Traffic Event Intelligence

Predictive Analytics for Bengaluru Traffic Police
Flipkart Grid 7.0 - Problem Statement 2

1. Problem Reframing

The dataset is labeled 'Rallies, Festivals & Traffic Management' but contains 8,173 rows of incident logs: 94% are unplanned disruptions (vehicle breakdowns, potholes, accidents), not rallies or festivals. Only 191 rows are true public events. We reframe honestly: every logged disruption is an 'event' consuming traffic-management resources, and we predict its impact and resourcing need.

Dataset: 7,944 clean events (Nov 2023 - Apr 2024), 23 corridors, 46 original columns expanded to 51 engineered features. Key challenge: 937 events (11.8%) are right-censored (still active at data snapshot).

2. Three Complementary Methods

Method A: Gradient-Boosted Classifiers

Two LightGBM classifiers predict event priority (high/low) and road closure (yes/no). Priority model achieves ROC-AUC 0.9995, F1 0.9993. Road closure model achieves PR-AUC 0.9946 despite 92:8 class imbalance. The near-perfect metrics are not overfitting - corridor_clean alone nearly determines priority because BTP has different operating rules per corridor.

Method B: Survival Analysis (Censoring-Aware)

Log-Normal Accelerated Failure Time model predicts event duration while correctly handling 937 right-censored events. C-index 0.7076, median absolute error 1.59h, calibration 75.3%. Key finding: vehicle breakdowns resolve in ~42 min while water logging and potholes often exceed the 24h observation window.

Method C: Spatio-Temporal Risk Forecasting

Corridor x time-slot grid (22 corridors x 6 daily slots) with LightGBM Poisson objective. Complete grid includes zero-event cells (critical for count data). Enables proactive deployment: 'Mysore Road 20:00-24:00 expects 2.1 events - pre-position 6 personnel.'

Method	Model	Primary Metric	Value
A: Classification	LightGBM	ROC-AUC (Priority)	0.9995
A: Classification	LightGBM	PR-AUC (Closure)	0.9946
B: Survival	Log-Normal AFT	C-index	0.7076
C: Risk Forecast	LightGBM Poisson	RMSE	1.510

3. Novelty Layers

Composite Impact Score (0-100)

No severity metric exists in the data. We construct a transparent weighted proxy: duration (30%), priority (20%), road

closure (20%), cause severity (15%), corridor importance (10%), rush hour (5%). Mean score: 43.0, max: 89.4. Four categories: Low (2%), Medium (79%), High (18%), Critical (0.8%).

DBSCAN Hotspot Discovery

38% of events (3,025) are off-grid. DBSCAN clustering (eps=500m, min_samples=5) reveals 79 unofficial hotspots. Notable: a 45-event tree_fall cluster near MSR Nagar suggests a specific vegetation/infrastructure issue BTP could prevent proactively.

Post-Event Learning Loop

The problem statement says 'no post-event learning system.' We build one using EMA risk tracking (alpha=0.1) and Bayesian Beta closure probability. Key insight: Bannerghata Road drifted +7.4 risk points (biggest riser), suggesting BTP should reallocate resources from improving corridors (ORR East 1: -5.3) to rising ones.

4. Key Findings & Recommendations

- **Deploy:** Pre-position on Mysore Road & Bannerghata Road during 20:00-24:00 (highest risk)
- **Investigate:** 45-event tree_fall cluster near MSR Nagar - vegetation/infrastructure investigation
- **Shift resources:** From ORR East 1 to Bannerghata Road based on risk drift analysis
- **Adopt:** The post-event learning loop for continuously updated risk estimates
- **Use:** The impact simulator for real-time resource allocation decisions

5. Stated Assumptions

- **Rule compliance:** No external data used - all analysis from provided dataset only
- **Resources:** Resource recommendations are calibrated heuristics, not learned from data
- **Timezone:** Timestamps are IST (confirmed by peak hour analysis)
- **Proxies:** Corridor importance proxied by event density (no traffic volume data)

6. Technical Stack & Deliverables

Deliverable	Technology	Purpose
pipeline.ipynb	Python/Jupyter	Code + documentation
dashboard.py	Streamlit/Plotly	Interactive 6-tab dashboard
react-dashboard/	React/Vite	Static results showcase
models/	LightGBM/lifelines	Saved models & metrics
Maps	MapMyIndia (Mappls)	Spatial visualization

Pipeline modules: data_loader.py, model_classification.py, model_survival.py, model_risk_forecast.py, impact_score.py, hotspot_discovery.py, post_event_learning.py. Total: ~2,500 lines of documented Python across 7 modules + 2 dashboard implementations.